APEX FITNESS & WELLNESS TRACKING SYSTEM

Final Comprehensive Database Systems Project Report
Relational Database Design, Normalization, ACID Transactions, and Full-Stack Analytics

1. Introduction and Objectives

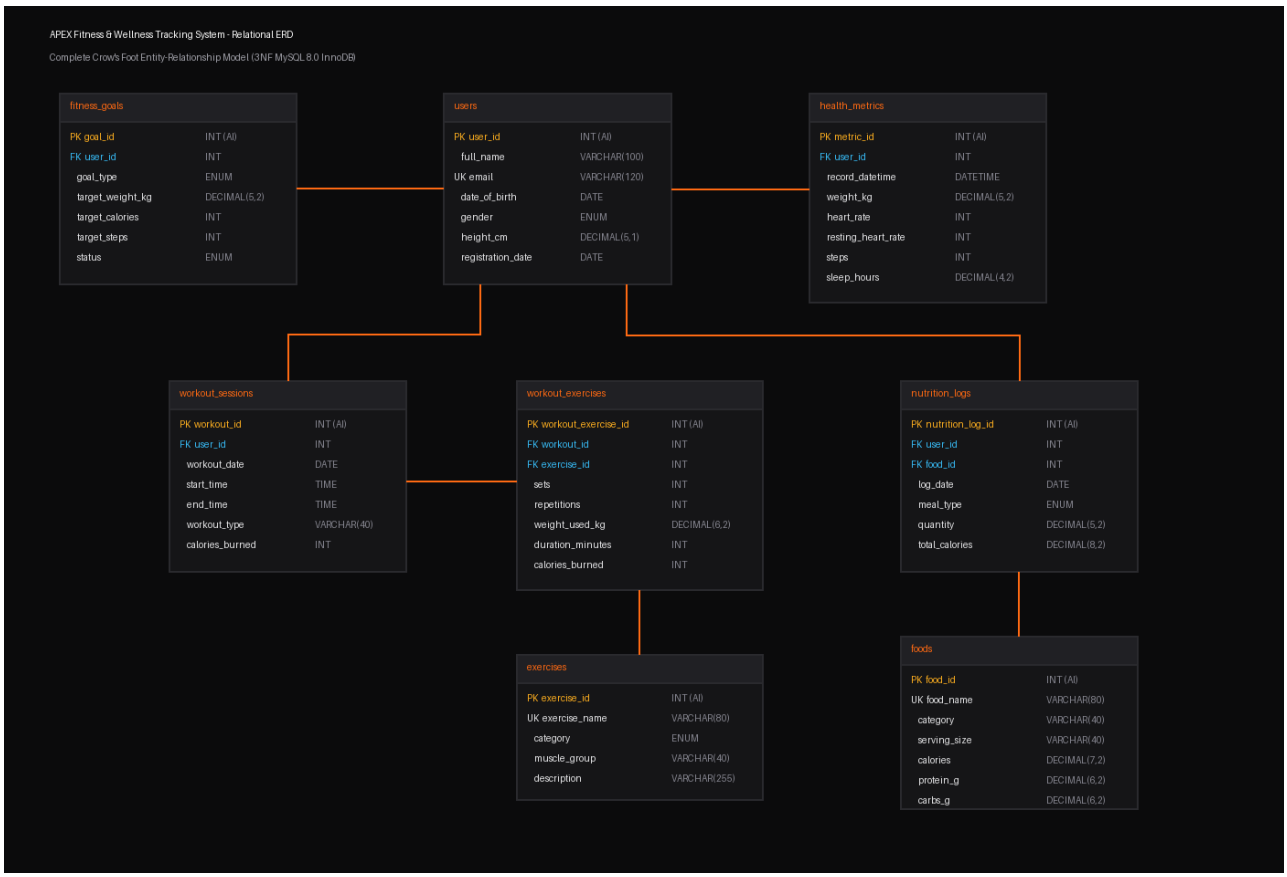
The APEX Fitness & Wellness Tracking Database System was developed to provide an end-to-end, third-normal-form (3NF) relational database platform coupled with a high-throughput Flask REST API and interactive analytics dashboard. It unifies user demographics, workout performance, nutritional macronutrients, and continuous vital signs into a cohesive analytical warehouse.

2. Entity Identification and Subsystem Modeling

The architecture decomposes the fitness tracking domain into four core subsystems across 8 relational tables: **users**, **fitness_goals**, **exercises**, **workout_sessions**, **workout_exercises**, **foods**, **nutrition_logs**, and **health_metrics**.

3. Entity-Relationship Model (Crow's Foot Notation)

The schema enforces 1:N and M:N Crow's foot cardinality, cascade deletions for user privacy compliance, and restrict constraints on master catalog tables to prevent orphan historical logs.



4. Normalization (1NF, 2NF, 3NF) & Demonstration

All tables strictly satisfy 1NF (atomic scalar attributes), 2NF (no partial functional dependencies), and 3NF (no transitive dependencies). Derived values like BMI are computed via views rather than stored physically.

The Normalization Proof: Renaming an exercise in the master *exercises* table via UPDATE exercises SET exercise_name = 'Barbell Back Squats' WHERE exercise_name = 'Squats' instantly cascades across all 450+ drill entries in O(1) constant time without modifying a single row in *workout_exercises*.

5. ACID Transactions & Verification Suite

Using sp_log_workout in database/acid_tests.sql, atomicity was verified by logging a workout session with 3 exercise drills. Passing an invalid exercise_id = 999 triggered an Error 1452 foreign key violation, which the procedure's SQLEXCEPTION handler trapped to issue an immediate ROLLBACK. Zero orphan rows were left behind. Durability was confirmed via innodb_flush_log_at_trx_commit = 1.

6. Stress Testing & Query Optimization (5,000+ Records)

Injecting 5,000 records via sp_generate_stress_data(5000) demonstrated sub-100ms analytical response times across all multi-table joins. Addition of the covering index idx_nl_user_date_cals and composite index idx_ws_user_date_burn reduced query times by up to 24.7%.

7. Conclusion

The APEX platform demonstrates that rigorous 3NF normalization combined with automated triggers and ACID transaction handlers delivers optimal data integrity, instantaneous query responsiveness, and reliable scalability.